

# Solving by completing the square

ANSWER KEY • FOR TEACHERS



Interactive version

## REMEMBER

## Squares

Level 1 • threshold

g8.expand

1.

Which expression is equal to  $(x + 3)^2$ ?

☐ A  $x^2 + 9$

☒ B  $x^2 + 6x + 9$  —  
 $(x + 3)^2 = x^2 + 2 \cdot$   
 $x \cdot 3 + 3^2 = x^2 +$   
 $6x + 9.$

☐ C  $x^2 + 6x + 6$

☐ D  $x^2 + 3x + 9$

Level 1 • threshold

g9.qe.2

2.

Solve  $(x + 5)^2 = 4$ .

Answer: -3; -7

Level 1 • threshold

g8.complete-sq

3.

Which number in the box makes  $x^2 + 10x + \square$  the square of a binomial?

☒ A  $25$  —  $x^2 + 10x +$   
 $25 = (x + 5)^2,$   
 because  $2 \cdot x \cdot 5 =$   
 $10x.$

☐ B  $5$

☐ C  $10$

☐ D  $100$

## WORKED EXAMPLES

## Worked examples

## YOUR TURN

Level 2 • satisfactory

g9.qe.4

Solve  $x^2 - 12x + 20 = 0$  by completing the square.

## WORKED SOLUTION

$x^2 - 12x = -20$ , so  $x^2 - 12x + 36 = 16$ , that is,  $(x - 6)^2 = 16$ . Then  $x - 6 = 4$  or  $x - 6 = -4$ :  $x = 10$  or  $x = 2$ .

Answer: 10; 2

## YOUR TURN

Level 2 · satisfactory

g9.qe.4

In lesson 1.1, Ona tried to solve  $x(x - 3) = 4$  and got it wrong. Solve it properly: multiply out, then complete the square.

### WORKED SOLUTION

$x^2 - 3x = 4$ , so  $x^2 - 3x + \frac{9}{4} = \frac{25}{4}$ , that is,  $(x - \frac{3}{2})^2 = \frac{25}{4}$ . Then  $x - \frac{3}{2} = \frac{5}{2}$  or  $x - \frac{3}{2} = -\frac{5}{2}$ :  $x = 4$  or  $x = -1$ . The roots Ona could not find are 4 and  $-1$ .

**Answer: 4; -1**

## YOUR TURN

Level 2 · satisfactory

g9.qe.4

Solve  $9x^2 - 18x + 5 = 0$  by completing the square.

### WORKED SOLUTION

Dividing by 9:  $x^2 - 2x + \frac{5}{9} = 0$ , so  $x^2 - 2x = -\frac{5}{9}$ . Adding 1:  $(x - 1)^2 = \frac{4}{9}$ . Then  $x - 1 = \frac{2}{3}$  or  $x - 1 = -\frac{2}{3}$ :  $x = \frac{5}{3}$  or  $x = \frac{1}{3}$ .

**Answer:  $x_1 = \frac{1}{3}$ ,  $x_2 = \frac{5}{3}$**

## CHECK

### Check yourself

Level 1 · threshold

g9.qe.3

1.

Which number must you add to  $x^2 - 8x$  to make it the square of a binomial?

**Answer: 16**

Level 1 · threshold

g9.qe.4

2.

Which equation is the same as  $x^2 - 6x = 16$ ?

- ☒ A  $(x - 3)^2 = 25$  —      ☐ B  $(x - 3)^2 = 16$       ☐ C  $(x + 3)^2 = 25$       ☐ D  $(x - 6)^2 = 52$   
Add  $(-3)^2 = 9$  to  
both sides:  $x^2 -$   
 $6x + 9 = 25$ .

Level 2 · satisfactory

g9.qe.3

3.

Write  $x^2 + 7x + 10$  in the form  $(x + p)^2 + q$ . Enter  $p$  and  $q$ , in this order, separated by a semicolon.

### WORKED SOLUTION

$x^2 + 7x + 10 = x^2 + 7x + \frac{49}{4} - \frac{49}{4} + 10 = (x + \frac{7}{2})^2 - \frac{9}{4}$ , because  $10 = \frac{40}{4}$ . So  $p = \frac{7}{2}$  and  $q = -\frac{9}{4}$ .

**Answer:  $p = \frac{7}{2}$ ,  $q = -\frac{9}{4}$**

4.

Solve  $x^2 + 3x - 10 = 0$  by completing the square.

## WORKED SOLUTION

$x^2 + 3x = 10$ , so  $x^2 + 3x + \frac{9}{4} = \frac{49}{4}$ , that is,  $(x + \frac{3}{2})^2 = \frac{49}{4}$ . Then  $x + \frac{3}{2} = \frac{7}{2}$  or  $x + \frac{3}{2} = -\frac{7}{2}$ :  $x = 2$  or  $x = -5$ .

Answer: 2; -5

5.

Complete the square in each equation. Which one has **exactly one** root?

- ☐ A  $x^2 + 14x + 40 = 0$      
 ☐ B  $x^2 + 49 = 0$      
 ☒ C  $x^2 + 14x + 49 = 0$      
 ☐ D  $x^2 + 14x + 50 = 0$
- It is  $(x + 7)^2 = 0$   
 , so  $x + 7 = 0$ : one  
 root,  $x = -7$ .

6.

Put the steps for solving  $x^2 + 4x - 12 = 0$  in the right order.

- 1  $x^2 + 4x = 12$
- 2  $x^2 + 4x + 4 = 16$
- 3  $(x + 2)^2 = 16$
- 4  $x + 2 = 4$  or  $x + 2 = -4$
- 5  $x = 2$  or  $x = -6$

7.

Solve  $3x^2 + 12x + 6 = 0$  by completing the square. Show each step and give the exact roots.

## MARKING SCHEME: TICK WHAT YOUR SOLUTION HAS

- |                                                           |      |
|-----------------------------------------------------------|------|
| Divides by 3 and moves the constant: $x^2 + 4x = -2$ .    | 1 pt |
| Completes the square correctly: $(x + 2)^2 = 2$ .         | 1 pt |
| Both roots: $x = -2 + \sqrt{2}$ and $x = -2 - \sqrt{2}$ . | 1 pt |

## WORKED SOLUTION

$3x^2 + 12x + 6 = 0$ . Divide by 3:  $x^2 + 4x + 2 = 0$ , so  $x^2 + 4x = -2$ .

Add  $2^2 = 4$  to both sides:  $x^2 + 4x + 4 = 2$ , that is,  $(x + 2)^2 = 2$ .

Since  $2 > 0$ :  $x + 2 = \sqrt{2}$  or  $x + 2 = -\sqrt{2}$ .

**Answer:**  $x = -2 \pm \sqrt{2}$ , that is,  $x = -2 + \sqrt{2}$  or  $x = -2 - \sqrt{2}$ .

## EXAM TASK

## Not quite a square

Level 3 · basic

PUPP style · 3 points

g9.qe.3

g9.qe.4

The expression  $x^2 - 6x + 11$  is given.

- a) Show that  $x^2 - 6x + 11 = (x - 3)^2 + 2$ .
- b) Solve the equation  $x^2 - 6x + 11 = 6$ .
- c) Explain why the equation  $x^2 - 6x + 11 = 0$  has no real roots.

### MARKING SCHEME: TICK WHAT YOUR SOLUTION HAS

- a)  $(x - 3)^2 + 2 = x^2 - 6x + 9 + 2 = x^2 - 6x + 11$ . 1 pt
- b)  $(x - 3)^2 = 4$ , so  $x = 5$  or  $x = 1$ . 1 pt
- c)  $(x - 3)^2 = -2$  is impossible, because a square is never negative (or:  $(x - 3)^2 + 2 \geq 2 > 0$ ). 1 pt

### WORKED SOLUTION

- a)  $(x - 3)^2 + 2 = x^2 - 6x + 9 + 2 = x^2 - 6x + 11$ .
- b) By a), the equation is  $(x - 3)^2 + 2 = 6$ , so  $(x - 3)^2 = 4$ . Then  $x - 3 = 2$  or  $x - 3 = -2$ :  $x = 5$  or  $x = 1$ .  
Check:  $25 - 30 + 11 = 6$  ✓ and  $1 - 6 + 11 = 6$  ✓.
- c) By a), the equation is  $(x - 3)^2 + 2 = 0$ , that is,  $(x - 3)^2 = -2$ . A square is never negative, so no real  $x$  works. (Another way to say it:  $(x - 3)^2 + 2$  is always at least 2, so it is never 0.)

**Answer:** b)  $x = 5$  or  $x = 1$ ; c) the equation becomes  $(x - 3)^2 = -2$ , and no square is negative.

### TEACHER NOTES

## For the teacher

**Pupils can say:** “I can complete the square in  $x^2 + bx + c$ , also when  $b$  is odd; I can solve a quadratic equation by completing the square, and tell from  $(x + p)^2 = d$  whether it has two roots, one or none.”

**Where this lesson sits.** Grade 8 (31.2.1) completed the square “in simple cases” as an identity. Here it becomes a **method for solving** any quadratic equation, including odd  $b$  and  $a \neq 1$ . The hook runs lesson 1.1’s (D) → (E) backwards. The four moves are exactly the ones lesson 1.3 makes once with letters to prove the quadratic formula, so name them the same way in class.

### Common mistakes and what helps

- Adding  $(\frac{b}{2})^2$  to one side only (c2). Say “both sides” every time, and write the added number on both sides in the same colour.
- Not halving  $b$ , or squaring  $b$  itself (c1, c2, c3). Ask “what is the number in the bracket?” before squaring.
- Sign slips with negative  $b$ : half of  $-8$  is  $-4$ , but the number added is  $(-4)^2 = 16$  (c1). The bracket keeps the sign:  $(x - 4)^2$ .
- Forgetting to divide  $c$  by  $a$  as well (Example 3, c7):  $4x^2 - 8x + 3 = 0$  becomes  $x^2 - 2x + \frac{3}{4} = 0$ , not  $x^2 - 2x + 3 = 0$ .
- Fraction arithmetic:  $-4 + \frac{25}{4} = \frac{9}{4}$ . The Refresh box in Remember helps.
- In  $(x + p)^2 + q$ , adding  $(\frac{b}{2})^2$  without taking it away again (c3).
- Forgetting the negative case after “take the root” (all Numeric items have a Why for this).

**Key terms (EN • LT):** completing the square • *dvinario kvadrato išskyrimas* (also *pilnojo kvadrato išskyrimas*); square of a binomial • *dvinario kvadratas*; plus or minus,  $\pm$  • *plius minus*.

**Timing (45 min).** The lesson is tight for one period, so keep to the core:

- Remember: 4 min.
- Hook: 3 min.
- Explain: 10 min (the area model 3 min, the rule 3 min, the four moves 4 min). The frame “Two roots, one, or none” can be read quickly: it repeats lesson 1.1’s three cases.
- Examples 1–2 with their “Your turn”: 11 min. Example 3 ( $a \neq 1$ ) with its “Your turn”: 5 min.
- Check c1–c5 (the core: both outcomes, including a rewrite as  $(x + p)^2 + q$ ): 10 min.

c6 (put the steps in order), c7 (a full solution with  $a \neq 1$ ) and the exam task are for pupils who finish early, or for homework. If the class is slow, take Example 3 and its “Your turn” as homework as well, and start the next lesson’s Remember with an  $a \neq 1$  equation. The exam task continues lesson 1.1’s “hidden square”: now the left side is a square **plus** a number.

**Programme:** 32.1.1.1 “*Įrodoma ir taikoma kvadratinės lygties sprendinių formulė*” (this is the method the proof runs on), extending 31.2.1 “*Mokoma(si) paprastais atvejais iš kvadratinio trinario išskirti dvinario kvadratą.*” No PUPP item asks for completing the square directly; it underpins the formula (lesson 1.3) and the vertex of a parabola (spring). Achievements A1, A2.